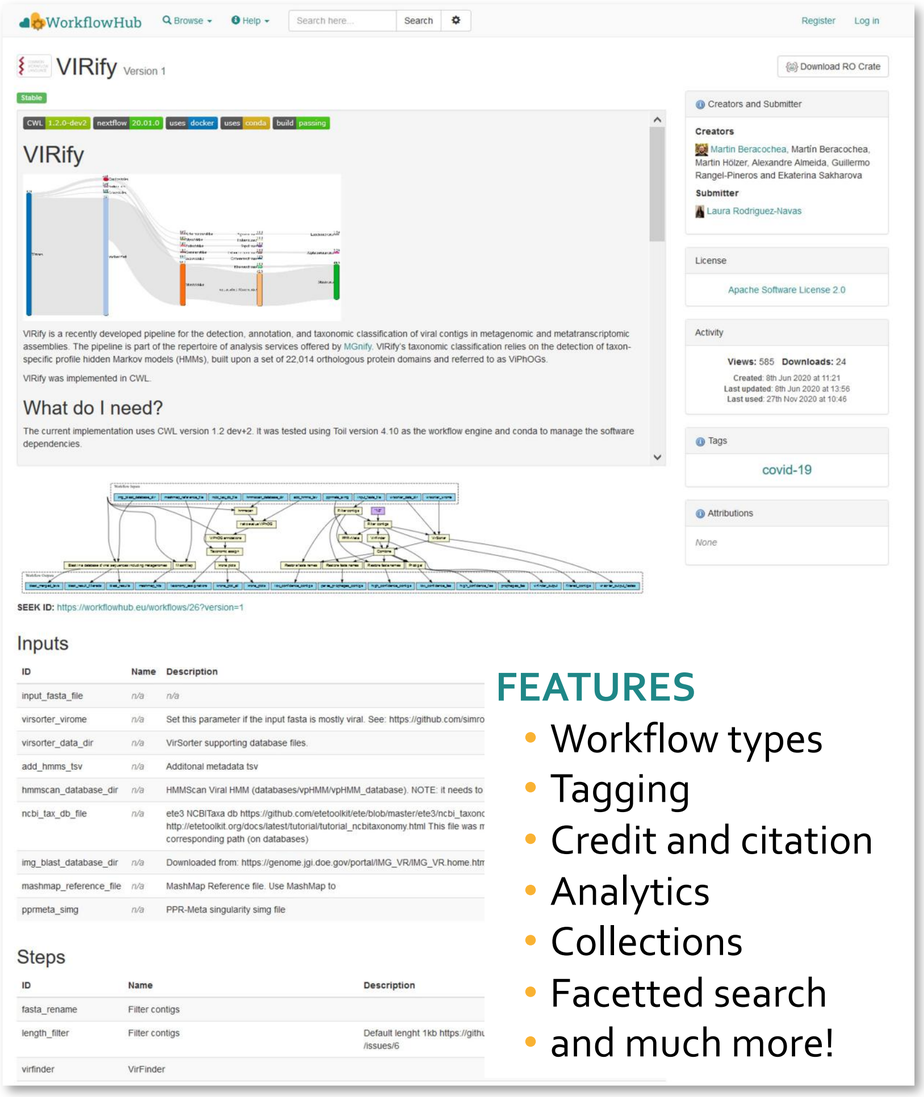
WorkflowHub

A FAIR registry for workflows

SCAN HERE
workflowhub.eu
workflowhub.org

Carole Goble[1], Finn Bacall[1], Stian Soiland-Reyes[1], Stuart Owen[1], Alan Williams[1], Ignacio Eguinoa[2], Bert Droesbeke[2], Hervé Ménager[3], Laura Rodriguez-Navas[4], José M. Fernández[4], Salvador Capella-Gutierrez[4], Michael R. Crusoe[5], Björn Grüning[6], Simone Leo[7], Luca Pireddu[7], Johan Gustafsson[8], Phil Ewels[9], WorkflowHub Community[10], and Frederik Coppens[2]

WorkflowHub is open to any workflows from any system in any discipline.



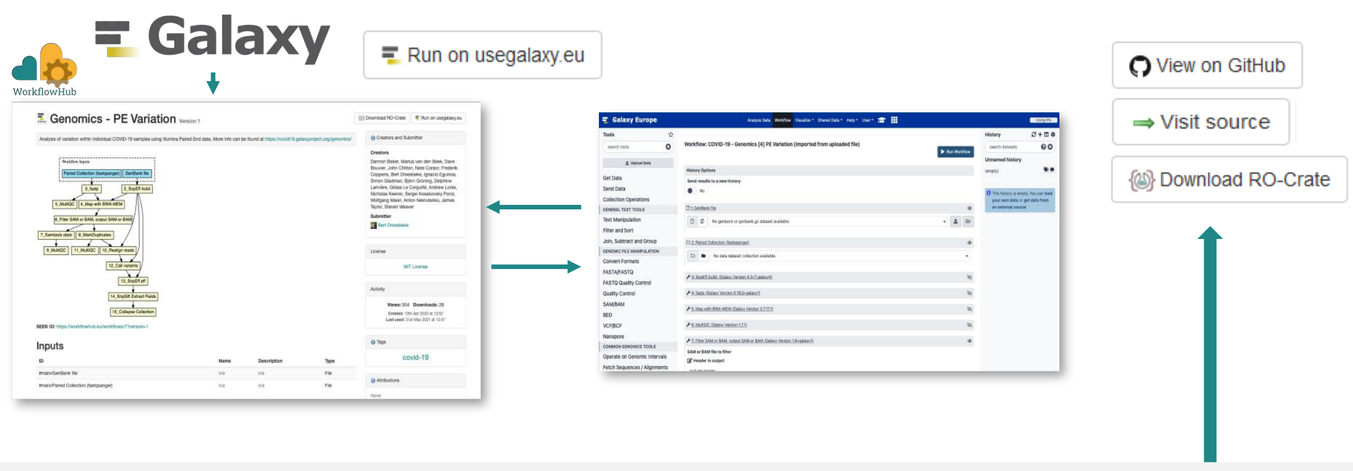
FEATURES

- Workflow types
- Tagging
- Credit and citation
- Analytics
- Collections
- Facetted search
- and much more!

The Hub is agnostic to Workflow Management Systems. We partner with popular and high profile platforms to co-develop and support metadata mark-up, packaging, registration, repository linkups, search & launching.

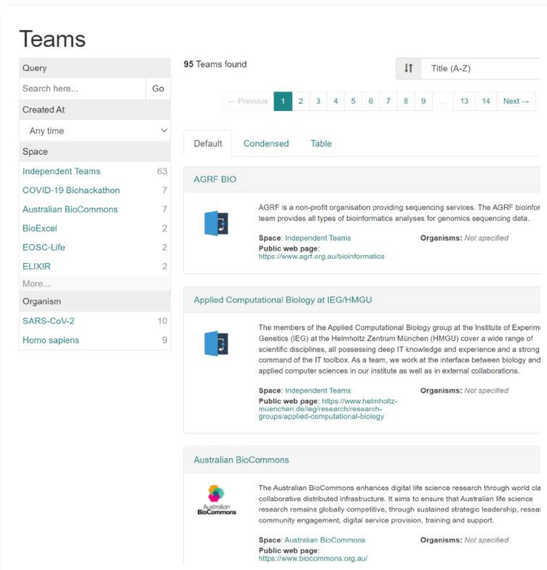
Galaxy · snakemake · nextflow · COMMON WORKFLOW LANGUAGE · jupyter

Workflows can remain and be accessed from their **native stores with Git support.** Discover workflows directly from a Workflow Management System (e.g. Galaxy Instance) or launch them from WorkflowHub using the **GA4GH Tool Registry Service API.**



Workflows can be uploaded / downloaded as Digital Research Objects using a specialised **RO-Crate profile**.

Workflows are treated as **publishable Digital Objects** with versioned identifiers & DOI assignments. They can be organised into collections to build canonical workflow libraries.



SPACES & TEAMS

Help organise workflow communities and build Communities of Practice. Links to discussion channels link users to workflow authors.

Workflow-RO-Crate is a format for *packaging* a workflow, its metadata and companion objects (links to containers, data etc) for archiving, reporting, citation; for exchanging workflows between workflow systems, execution platforms, testing & monitoring services; and for depositing in data repositories like DataVerse and Zenodo.

OPENEBENCH · Life Monitor · WfEx · SCHEMA · zenodo

A workflow metadata standards framework describes various aspects of the workflow for registration in the Hub and the RO-Crate, links to associated objects.

COMMON WORKFLOW LANGUAGE — Canonical workflow description of the *steps of the workflow*, including Abstract **CWL** generated by WfMS such as Galaxy.

EDAM — Type the inputs, outputs & steps using **EDAM ontology** annotations, the only bio-specific part.

Bioschemas.org / schema.org — **Schema.org** based common metadata *about the workflow, tools & parameters.*

JOIN THE WORKFLOWHUB CLUB

An **open community** that works together to perpetually develop the Hub and help its users. https://about.workflowhub.eu/

The WorkflowHub uses the open source **FAIRDOM-SEEK infrastructure** [2]. We acknowledge the support of COVID-19 Virtual Biohackathon2020, BCC2020 CoFest and Biohackathon Europe 2020.

REFERENCES

[1] Goble C et al FAIR computational workflows. Data Intelligence 2(2020), 108–121. doi: 10.1162/dint_a_00033
[2] Wolstencroft K et al SEEK: a systems biology data and model management platform, BMC Systems Biology, 9:33, 2015. doi:10.1186/s12918-015-0174-y.

AFFILIATIONS

[1]The University of Manchester, [2]VIB-UGent Center for Plant Systems Biology, [3]Institut Pasteur, [4]Barcelona Supercomputing Center, [5]Vrije Universiteit Amsterdam, [6]University of Freiburg, [7]CRS4, [8]Australian BioCommons, [9]Seqera Labs, [10]https://about.workflowhub.eu/acknowledgements/#workflowhub-club

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EOSC-Life · elixir · IBISBA · FAIRDOM · workflows community initiative · SYNTHESYS+ · bioexcel · Australian BioCommons